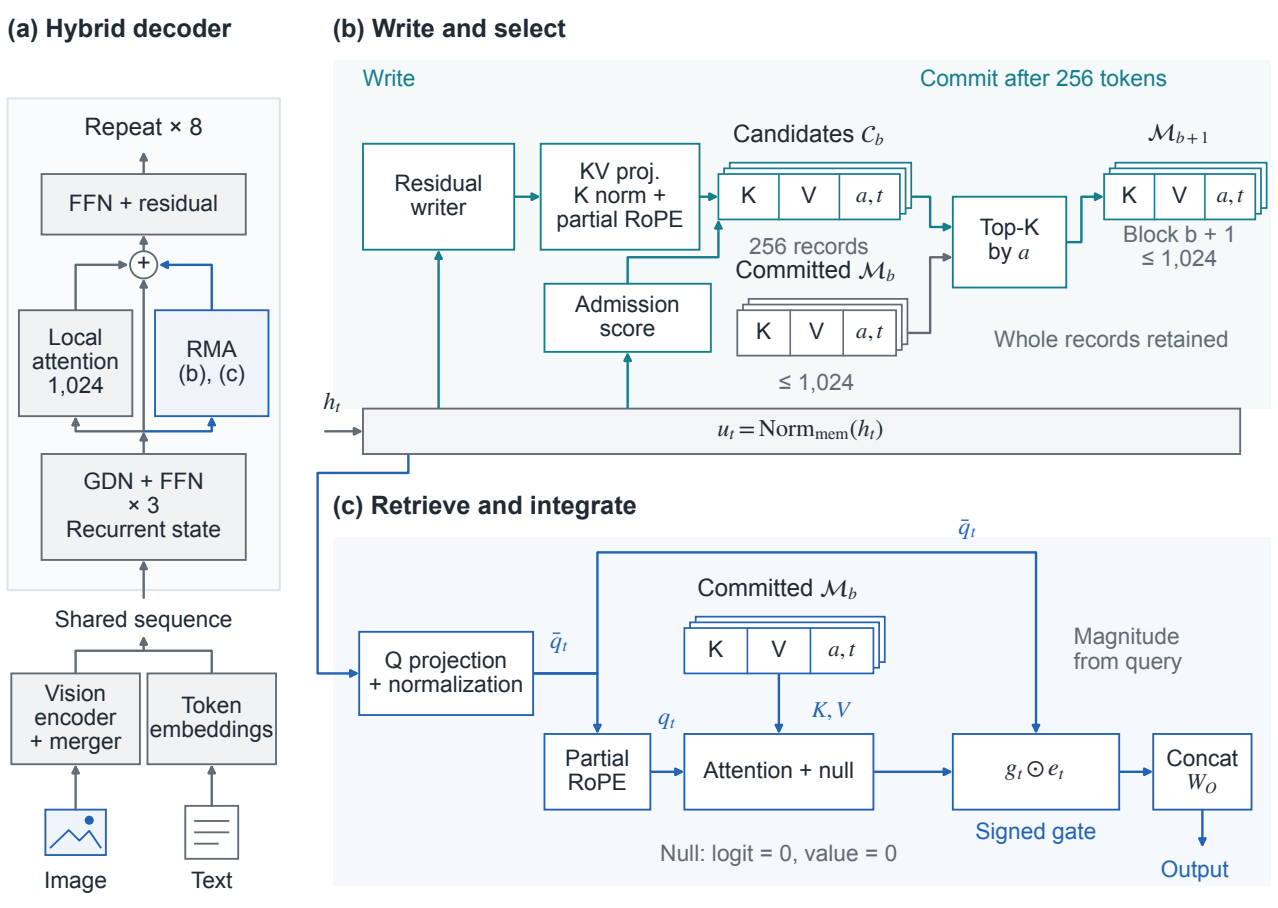

(a) Hybrid decoder
Repeat × 8
FFN + residual
Local attention 1,024
RMA (b), (c)
GDN + FFN × 3 Recurrent state
Shared sequence
Vision encoder + merger
Token embeddings
Image
Text
(b) Write and select
Write
Commit after 256 tokens
Residual writer
KV proj. K norm + partial RoPE
Candidates $\mathcal{C}_b$
K V a,t
256 records
Committed $\mathcal{M}_b$
Admission score
Top-K by a
$\mathcal{M}_{b+1}$
Block b + 1 ≤ 1,024
Whole records retained
≤ 1,024
h_t
$u_t = \mathrm{Norm}_{\mathrm{mem}}(h_t)$
(c) Retrieve and integrate
$\bar{q}_t$
Committed $\mathcal{M}_b$
Q projection + normalization
Magnitude from query
q_t
K, V
Partial RoPE
Attention + null
$g_t \odot e_t$
Concat W_O
Signed gate
Null: logit = 0, value = 0
Output